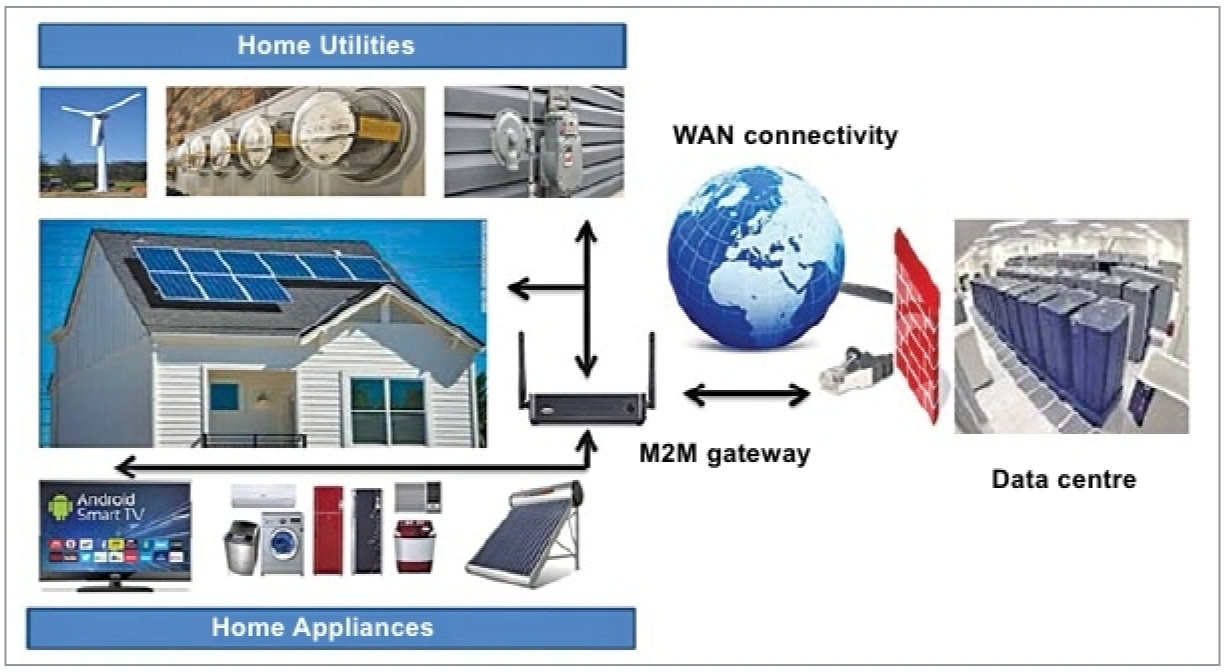

Fig. 2: Concept of the IoT explained using a smart metering system

3GPP PORTFOLIO OF TECHNOLOGIES AVAILABLE FOR THE IoT

	eMTC (LTE Cat M1)	NB-IoT	EC-GSM-IoT
Deployment	In-band LTE	In-band and guard-band LTE, standalone	In-band GSM
Coverage*	155.7dB	164dB for standalone, FFS others	164dB, with 33dBm power class and 154dB, with 23dBm power class
Downlink	OFDMA, 15kHz tone spacing, turbo code, 16 QAM, 1Rx	OFDMA, 15kHz tone spacing, TBCC, 1Rx	TDMA/FDMA, GMSK and 8PSK (optional), 1Rx
Uplink	SC-FDMA, 15kHz tone spacing, turbo code, 16 QAM	Single tone, 15kHz and 3.75kHz spacing, SC-FDMA, 15kHz tone spacing, turbo code	TDMA/FDMA, GMSK and 8PSK (optional)
Bandwidth	1.08MHz	180kHz	200kHz per channel. Typical system bandwidth of 2.4MHz (smaller bandwidth down to 600kHz being studied within Rel-13)
Peak rate (DL/UL)	1Mbps for both DL and UL	DL: ~250kbps UL: ~250kbps for multi-tone, ~20kbps for single-tone	For DL and UL (using four timeslots): ~70kbps (GMSK), ~240kbps (8PSK)
Duplexing	FD and HD (type B), FDD and TDD	HD (type B), FDD	HD, FDD
Power saving	PSM, ext. I-DRX, C-DRX	PSM, ext. I-DRX, C-DRX	PSM, ext. I-DRX
Power class	23dBm, 20dBm	23dBm, others TBD	33dBm, 23dBm

*In terms of MCL target. Targets for different technologies are based on somewhat different link budget assumptions

However, GSMA has stated that NB-IoT and LTE-M will both be essential components of 5G in connecting the IoT. It has urged mobile operators to offer services and products—not just connectivity—if they want to capture the US$ 1.1 trillion market opportunity by 2025.

3GPP standards for the IoT are also known as Cellular-IoT (C-IoT). Given below are the 3GPP technologies available for the IoT.

eMTC. This is an enhanced machine-type communication with new power-saving mode (PSM) for Cat 0 devices. 3GPP started working on this release 12 onwards, with a focus on providing low-data and low-power devices/sensors for long life sustainability and zero manual touch. It goes with LTE services and can cater to high data rates. It is flexible and focused enough for narrow-band uses like 1.08MHz.

NB-IoT. This is about specific changes in LTE standards, such as finding a tweak around LTE radio frame structure to support IoT data traffic, massive devices/sensors connectivity, widespread coverage and low-cost focus. It supports narrow-band usage up to 180kHz.

NB-IoT has three modes of operation, as follows:

1. Standalone mode, where there is a standalone spectrum for IoT services; spectrums mostly used for GSM/GPRS
2. Guard-band mode, which utilises LTE guard band's unused resources
3. In-band, which utilises resources from normal LTE carriers; also known as LTE-M

EC-GSM-IoT. This is about using GSM/GPRS technology for IoT devices/sensors connectivity. It also provides PSM enhancements and GSM security. EFY